UFUG 2104
Written Assignment 1

Jincan LI

March 9, 2026

Tips: Please clilck the link below to get to the targeted answers.

Problem 1
Problem 2
Problem 3
Problem 4
Problem 5

1 Problem 1

(16 pts) The pdf (probability density function) of X is given by

$$f_X(x) = \begin{cases} -\frac{1}{4}x, & -2 \le x < 0 \\ cx^2, & 0 \le x \le 3 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of the constant c.

(b) Plot the pdf of X from -3 to 4. If you plot by hand, make sure that your plot shows the main features, i.e., where they intercept the y-axis and x-axis. (c) Find the expression for the cdf (cumulative distribution function) of X. (d) Find the expected value of X. (e) Find the probability of the event $A = \{1 \le x \le 2\}$. (f) Find the conditional pdf of X given A, $f_{X|A}(x|A)$ and plot it from -3 to 4. If you plot by hand, make sure that your plot shows the main features, i.e. where they intercept the y- axis and x-axis. (g) Find the conditional expected value of X given A, $\mathbb{E}[X|A]$

Solution

(a) By the definition of pdf, we have

$$\int_{-\infty}^{\infty} f_X(x)dx = 1$$

and given the property of the function, we have

$$\int_{-\infty}^{\infty} f_X(x)dx = \int_{-\infty}^{-2} 0dx + \int_{-2}^{0} \left(-\frac{1}{4}x\right) dx + \int_{0}^{3} (cx^2)dx + \int_{4}^{\infty} 0dx = 1$$

Which means

$$(-\frac{1}{8}x^2)\Big|_{-2}^{0} + (\frac{1}{3}cx^3)\Big|_{0}^{3} = \frac{1}{2} + 9c = 1$$

$\Rightarrow \boxed{c = \frac{1}{18}}$

(b) The pdf can be graphed as below:

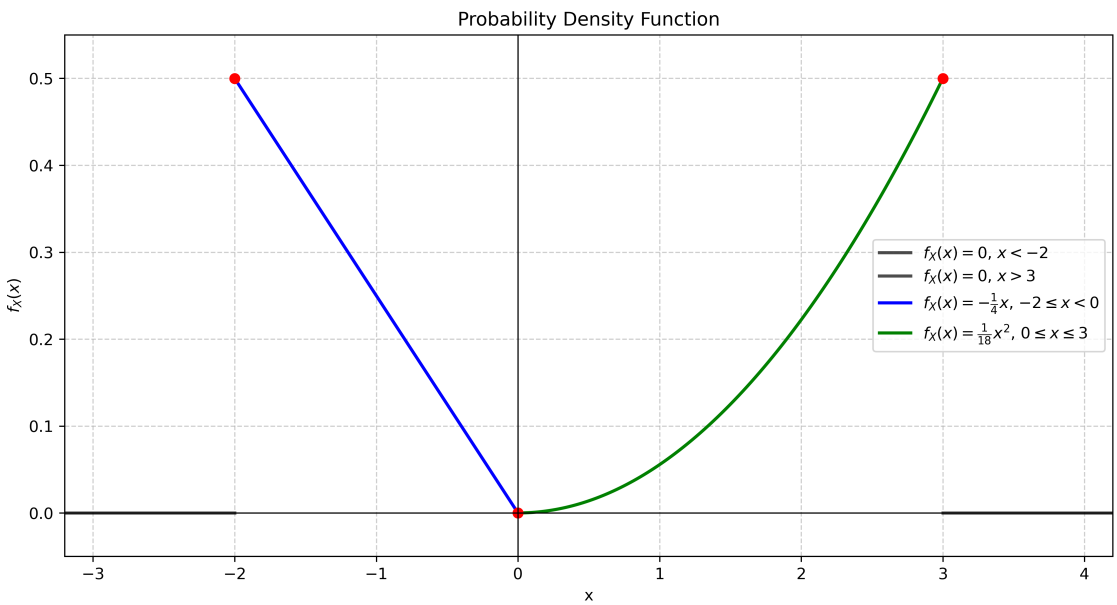

Figure 1

(c) The cdf can be calculated as below:

$$F_X(x) = \begin{cases} 0 & x < -2 \\ \frac{1}{2} - \frac{1}{8}x^2 & -2 \leq x < 0 \\ \frac{1}{2} + \frac{1}{54}x^3 & 0 \leq x \leq 3 \\ 1 & 3 < x \end{cases}$$

2 Problem 2

3 Problem 3

4 Problem 4

5 Problem 5